Assignment

Chapter 6: Memory Management and Virtual Memory

- **7.12** Consider a simple paging system with the following parameters: 2^{32} bytes of physical memory; page size of 2^{10} bytes; 2^{16} pages of logical address space
  - a. How many bits are in a logical address?
  - b. How many bytes in a frame?
  - c.How many bits in the physical address specify the frame?
  - d. How many entries in the page table?
  - e. How many bits in each page table entry? Assume each page table entry contains a valid/invalid bit.

Chapter 6

- **8.1** Suppose the page table for the process currently executing on the processor looks like the following. All numbers are decimal, everything is numbered starting from zero, and all addresses are memory byte addresses. The page size is 1,024 bytes.

Virtual page number	Valid bit	Reference bit	Modify bit	Page frame number
0	1	1	0	4
1	1	1	1	7
2	0	0	0	—
3	1	0	0	2
4	0	0	0	—
5	1	0	1	0

Chapter 6

- b. What physical address, if any, would each of the following virtual addresses correspond to?(Do not try to handle any page faultsif any):
- (i) 1052
- (ii) 2221
- (iii) 5499

Chapter 6

- **8.4** Consider the following string of page references 7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2. Complete a figure similar to Figure 8.15, showing the frame allocation for:
  - a.FIFO(first-in-first-out)
  - b.LRU(least recently used)
  - c.Clock
  - d.Optimal (assume the page reference string continues with 1,2, 0, 1, 7, 0, 1)
  - e.List the total number of page faults and the miss rate for each policy. Count page faults only after all frames have been initialized.

Chapter 6

- **8.17 (Seventh Edition)** Assume that a task is divided into four equal-sized segments and that the system builds an eight-entry page descriptor table for each segment. Thus, the system has a combination of segmentation and paging. Assume also that the page size is 2 Kbytes.
  - a.What is the maximum size of each segment?
  - b.What is the maximum logical address space for the task?
  - c. Assume that an element in physical location 00021ABC is accessed by this task. What is the format of the logical address that the task generates for it? What is the maximum physical address space for the system?

Assignment

Chapter 7: Uniprocessor Scheduling

- **9.2** Consider the following set of processes

process	Arrival Time	Processing Time
A	0	3
B	1	5
C	3	2
D	9	5
E	12	5

Perform the same analysis as depicted in Table 9.5 and Figure 9.5 for this set(RR,FB/q=1).